



Design and Development of a Web-Based Internship and Placement Management System Using Java

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA0927 – Programming in Java
to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

Computer Science and Engineering

Submitted by

N.Tejaswitha (192371057)

N.Vijetha (192311258)

B.Neha (192425033)

V.S.Meenakshy (192325133)

Under the Supervision of

Dr.S.Pushpa Latha

Professor / CSE

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Design and Development of a Web-Based Internship and Placement Management System Using Java**” has been carried out by **N. Tejaswitha (192371057), N.Vijetha (192311258), B.Neha (192425033), V. S. Meenakshy (192325133)** under the supervision of **Dr. S. Pushpa Latha** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer Science & Engineering program** at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.Charlyn Pushpalatha

Professor

Department of CSE

SIMATS

SIMATS , Chennai - 602105.

COURSE FACULTY

Dr.S.Pushpalatha

Professor

Department of CSE

SIMATS

SIMATS , Chennai - 602105.

Submitted for the Project work Viva-Voce held on_____

INTERNAL EXAMINER

EXTERNAL EXAMINER

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

DECLARATION

We, **N.Tejaswitha (192371057), N.Vijetha (192311258), B.Neha (192425033), V.S. Meenakshy (192325133)** of the **B.Tech Computer Science & Engineering program** , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Design and Development of a Web-Based Internship and Placement Management System Using Java**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:

Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
N. Vijetha (192311258)	Student Management Module	Student registration, login, profile, resume upload, application features	Tested student functionalities	Student module documenta ti on	25%
N. Tejaswitha (192371057)	Company Management Module	Company registration, profile, job posting, application management	Tested company operations	Company module documentatio n	25%
V. S. Meenakshy (192325133)	Internship & Placement Management Module	Opportunity management, application processing, status tracking	Tested internship and placement workflow	Module description and results	25%
B. Neha (192425033)	Admin and System Management Module	Admin login, user management, system control	Tested admin functions and database operations	Conclusion and final report preparation	25%
Total					100%

ABSTRACT

Managing internship and placement activities is a challenging process for educational institutions due to the involvement of multiple stakeholders, including students, companies, and admin. Traditional methods of handling recruitment activities rely on manual records, spreadsheets, and separate communication channels, which may lead to difficulties in maintaining data accuracy, tracking applications, managing opportunities, and monitoring recruitment progress. The absence of a centralized platform reduces efficiency and creates challenges in organizing internship and placement information. The significance of this project is to provide an automated and structured solution that simplifies the internship and placement management process. The proposed **Web-Based Internship and Placement Management System Using Java** provides a centralized platform that connects students, companies, and admin to efficiently manage recruitment activities. The system is developed using **Java, JSP, Servlets, JDBC, HTML, CSS, and MySQL**. The methodology includes requirement analysis, system design, database design, module development, integration, and testing. The system consists of major modules such as Student Management, Company Management, Internship and Placement Management, and Admin & System Management. Students can register, manage profiles, upload resumes, view opportunities, apply for internships and placements, and track application status. Companies can create profiles, post opportunities, view applications, and manage candidate selection. Admin can monitor and manage overall system activities. The implementation results show that the system successfully automates major internship and placement processes, reduces manual effort, improves data organization, and provides efficient management of recruitment records. The system ensures better coordination between students and companies through a centralized database-driven platform. The principal conclusion of this project is that the developed system provides an effective and reliable solution for managing internship and placement activities. It improves efficiency, reduces paperwork, and simplifies the recruitment workflow. Future enhancements can include mobile application support, AI-based recommendations, real-time notifications, and advanced analytics features.

Keywords: Internship Management, Placement Management, Java Web Application, Student Portal, Recruitment System, MySQL Database.

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	02
ii	DECLARATION BY THE CANDIDATE	03
iii	INDIVIDUAL CONTRIBUTION STATEMENT	04
iv	ABSTRACT	05
v	TABLE OF CONTENTS	06
vi	LIST OF FIGURES	07
vii	LIST OF TABLES	08
1	CHAPTER 1: Introduction and Engineering Problem	09-12
2	CHAPTER 2: Literature Review	13-15
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	16-20
4	CHAPTER 4: Implementation, Testing & Results, Project Management	21-30
5	CHAPTER 5: Conclusion, Future Work and Reflection	31-32
	References	33
	Appendices	34-41

LIST OF FIGURES

Figure No	Figure Name	Page No
FIG 01	Student Management Module	12
FIG 02	Company Management Module	15
FIG 03	Eclipse IDE Implementation	22
FIG 04	MySQL Workbench	22
FIG 05	Internship & Placement Portal Home Page	24
FIG 06	Student Registration Page	25
FIG 07	Student Dashboard	25
FIG 08	Company Registration	25
FIG 09	Company Dashboard	26
FIG 10	Admin Login Page	26
FIG 11	Admin Dashboard	27
FIG 12	System Architecture of Internship and Placement Management System	34
FIG 13	Database Design of the Proposed System	34
FIG 14	Web Application Development and Implementation Using Eclipse IDE	35
FIG 15	Database Design and Implementation Using MySQL	35
FIG 16	Internship & Placement Portal	36
FIG 17	Student Dashboard	36
FIG 18	Company Dashboard	37
FIG 19	Admin Dashboard	37
FIG 20	Server Gets Started	38
FIG 21	Company Registration	39
FIG 22	Company Login	39
FIG 23	Company Dashboard	40
FIG 24	Post Internship	40
FIG 25	View Posted Internships	40
FIG 26	View Internship Applications	41
FIG 27	Post Placement	41
FIG 28	View Placement Applications	41

LIST OF TABLES

Table No	Table Name	Page No
Table 3.1	Stakeholder / User Requirements	16
Table 3.1	Functional & Non-Functional Requirements	16
Table 3.2	Constraints & Applicable Standards	17
Table 3.3	Design Specifications	17
Table 3.4	Alternative Solutions & Evaluation	18
Table 3.6	Trade-off Analysis	19
Table 3.7	Sustainability, Ethics, Safety & Security Analysis	20
Table 3.7	Risk Register	20
Table 4.1	Development Resources Used	21
Table 4.2	Tools / Software / Equipment Used	22
Table 4.3	Test Plan	23
Table 4.4	Verification Results	23
Table 4.5	Major Results of Implemented Modules	24
Table 4.6	Comparison with Existing Solutions & Achievement of Objectives	28
Table 4.8	Project Timeline / Milestone Plan	29
Table 4.8	Student Contribution	30
Table 4.8	Project Budget	30
Table D.1	System Testing Results	38

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Internship and placement management is an important process in educational institutions for connecting students with career opportunities. The process involves managing student details, company information, internship and placement opportunities, applications, and recruitment activities. Traditional methods depend on manual records and separate communication channels, making it difficult to maintain accurate data and track application progress.

The **Web-Based Internship and Placement Management System Using Java** provides a digital platform to automate and organize these activities. The system uses Java-based web technologies with database integration to provide efficient management of internship and placement processes. It enables interaction between students, companies, and admin through a centralized system.

1.2 Need for the Project

Internship and placement activities involve large amounts of information that need proper management and continuous monitoring. Manual handling of student records, company details, and applications can result in errors, delays, and difficulty in tracking recruitment progress.

The problem affects students, companies, and admin involved in the recruitment process. Students may face difficulties in finding opportunities and tracking applications, while companies may find it challenging to manage candidate information. Admin faces difficulties in maintaining and organizing recruitment records.

Existing systems have limitations such as lack of centralized data management, limited application tracking, and increased dependency on manual processes. The engineering

significance of this project is to develop an automated software solution that improves efficiency, data organization, and communication between stakeholders.

1.3 Problem Statement

Educational institutions require an efficient system to manage internship and placement activities involving multiple users such as students, companies, and admin. The existing manual approach creates challenges in maintaining records, managing opportunities, tracking applications, and updating recruitment status.

The engineering problem is to design and develop a centralized web-based system that can handle multiple interacting modules, provide secure user access, maintain accurate database records, and simplify internship and placement management within realistic software development constraints.

1.4 Project Aim

The aim of this project is to develop a **Web-Based Internship and Placement Management System Using Java** that automates internship and placement activities by providing a centralized platform for students, companies, and admin.

1.5 Project Objectives

1. To design a web-based platform for managing internship and placement activities.
2. To develop modules for student, company, internship, placement, and admin management.
3. To implement secure registration and login functionalities for different users.
4. To integrate database connectivity using JDBC and MySQL for efficient data storage.
5. To test the system functionalities and ensure reliable performance.

6. To evaluate the effectiveness of the system in reducing manual effort and improving management efficiency.

1.6 Scope

Included:

- Student registration, profile management, and resume upload.
- Company registration and opportunity posting.
- Internship and placement application management.
- Application status tracking.
- Admin management of users and system records.
- Database storage and retrieval of information.

Excluded:

- Mobile application development.
- AI-based job recommendations.
- Real-time notification services.
- Advanced analytics features.

Assumptions:

- Users have valid login credentials.
- Companies provide accurate opportunity details.
- The system operates within a web-based environment.

Boundary Conditions:

- The system is designed for institutional internship and placement management.
- Access is limited to registered students, companies, and admin.

1.7 Expected Engineering Outcome

The expected outcome of this project is a **web-based software system** that provides an organized platform for managing internship and placement activities. The system enables efficient interaction between students, companies, and admin through different modules.

The developed solution provides:

- A centralized database-driven management system.
- Automated handling of student applications and company opportunities.
- Improved recruitment process management.
- Reduced paperwork and manual effort.
- A reliable prototype for internship and placement administration.

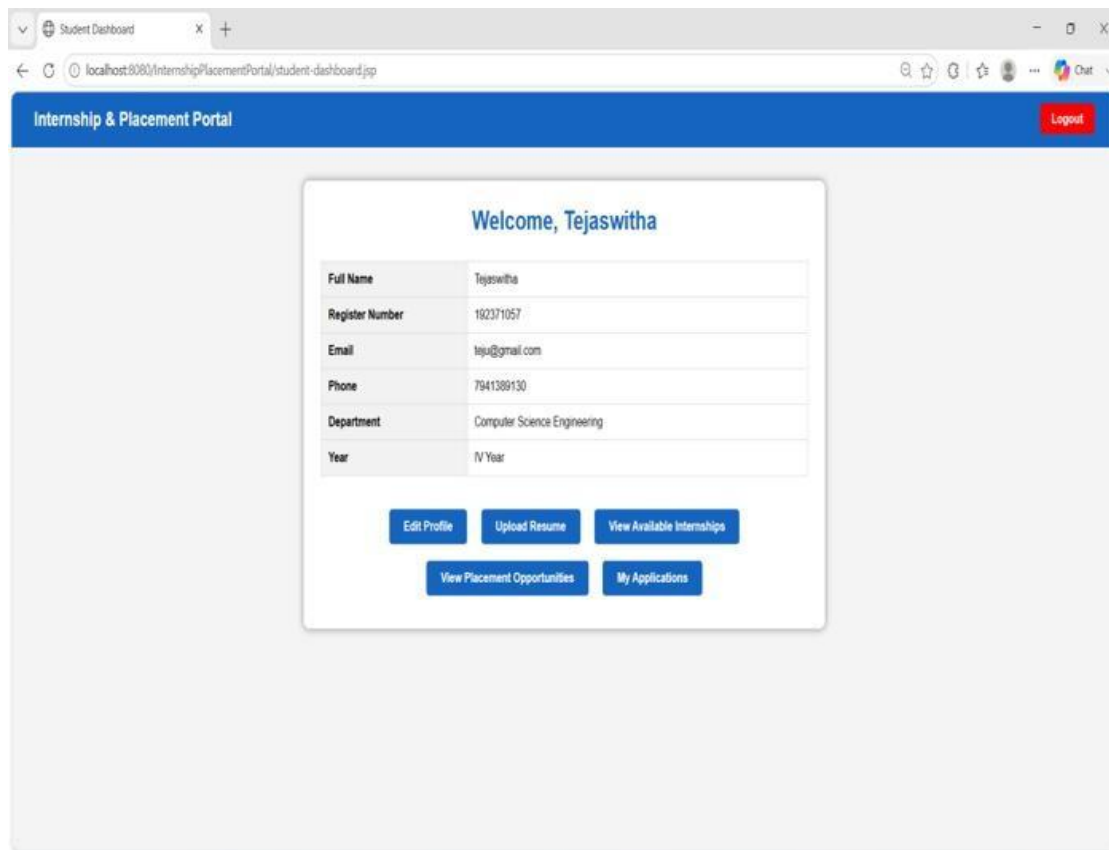


Fig. 01 Student Management Module

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted by studying existing internship and placement management systems, web-based recruitment platforms, research papers related to digital recruitment solutions, and technologies used for developing enterprise web applications. Relevant information was identified through analysis of existing software solutions, database management approaches, web application architectures, and Java-based development frameworks. The review focused on understanding current limitations, available technologies, and possible improvements required for developing an efficient internship and placement management system.

2.2 Review of Existing Work

Existing internship and placement management systems mainly focus on automating student registration, company recruitment, job posting, and application tracking processes. Many web-based solutions provide separate portals for students and recruiters to manage recruitment activities.

Research works on digital recruitment platforms highlight the importance of centralized databases, automated application processing, and role-based access control for improving recruitment efficiency. Technologies such as Java, JSP, Servlets, JDBC, MySQL, and MVC architecture are widely used for developing secure and maintainable web applications.

Commercial recruitment platforms provide features such as job searching, candidate management, and employer portals. However, these systems are generally designed for large-scale employment markets and may not completely satisfy institution-specific internship and placement requirements.

Existing methods provide basic recruitment management features, but challenges remain in providing a simple, customized, and integrated platform for students, companies, and admin within an academic environment.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Internship and Placement Process	Simple to implement and requires minimal technology	Time-consuming, paper-based, difficult tracking, higher chances of errors	Proposed system replaces manual work with automation
Spreadsheet-Based Management	Easy data entry and basic record maintenance	Lack of centralized access, poor security, difficult updates	Proposed system provides database-based management
General Recruitment Platforms	Provides job posting and candidate management features	Not customized for academic internship and placement activities	Proposed system focuses on institutional recruitment needs
Existing Web-Based Placement Systems	Automates recruitment activities and improves accessibility	May lack integrated modules and customization	Proposed system combines student, company, internship, placement, and admin modules

2.4 Research / Engineering Gap

Existing approaches still have limitations in providing a complete and customized solution for academic internship and placement management. Many systems do not provide an integrated platform that supports students, companies, and admin with equal functionality.

The major gaps identified are:

- Lack of centralized management of internship and placement activities.

- Difficulty in tracking application progress efficiently.
- Limited customization according to institutional requirements.
- Poor coordination between students, companies, and admin.
- Dependency on manual processes for maintaining recruitment records.

2.5 Proposed Contribution

The proposed **Web-Based Internship and Placement Management System Using Java** addresses these limitations by providing a centralized and structured platform for managing recruitment activities. The system integrates student management, company management, internship and placement management, and admin control modules.

The major contributions of the project include:

- Automated management of student and company records.
- Efficient internship and placement opportunity management.
- Online application submission and status tracking.
- Secure database storage using MySQL.
- Improved communication and coordination between stakeholders.
- Reduced paperwork and manual effort through digital automation.

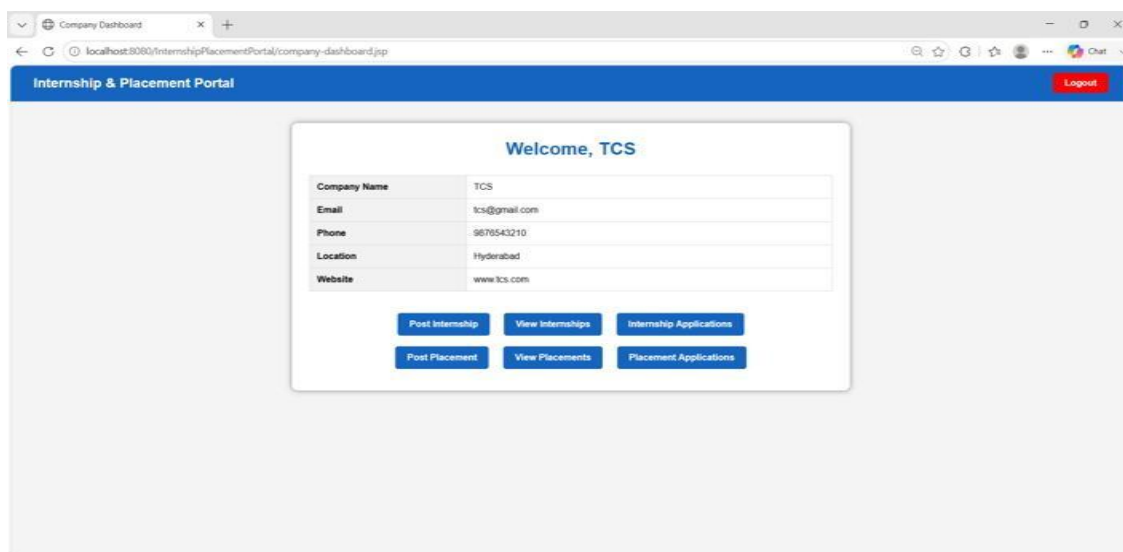


Fig. 02 Company Management Module

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Students	Register, manage profile, upload resume, view internships/placements, apply and track application status
Companies	Register, create company profile, post opportunities, view applications, update candidate status
Admin	Manage users, monitor internship and placement activities, maintain system records

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
User Registration and Login	Functional	Secure authentication for all users
Internship and Placement Management	Functional	Create, view and manage opportunities
Application Tracking	Functional	Real-time application status updates
Database Management	Functional	Store and retrieve records accurately
System Performance	Non-Functional	Response time below 3 seconds
Security	Non-Functional	Authorized access through login control

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
Java Programming Standards	Maintain structured and reusable code	Used Java, JSP and Servlets with modular programming
Database Management Principles	Ensure reliable data storage	MySQL database used for storing user and application data
Web Security Practices	Protect user information	Implemented authentication and controlled access
Software Testing Practices	Ensure system reliability	Functional testing performed for all modules

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
User Authentication	Secure login for students, companies and admin	Access Control	High
Database Storage	Efficient storage of recruitment records	Database	High
Application Processing	Manage internship and placement applications	Records	High
User Interface	Simple and user-friendly interface	Design	Medium
System Performance	Fast response and reliable operation	Seconds	High

3.4 Alternative Solutions & Evaluation

Alternative 1:

A manual internship and placement management approach using paper records and spreadsheets.

Alternative 2:

A general recruitment platform adapted for college placement activities.

Alternative 3:

A customized Java-based web application developed specifically for institutional internship and placement management.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
User Authentication	High	Low	Medium	High
Database Storage	Medium	Low	High	Medium
Application Processing	High	Low	Medium	High
User Interface	Medium	Low	Medium	High
System Performance	High	Low	Medium	High

3.5 Selected Design & Detailed Design

Justification for Final Design Selection

The Java-based web application was selected because it provides better customization, centralized database management, and efficient communication between students, companies,

and administrators. The modular architecture improves maintainability and allows future enhancements.

System Architecture

Student Module → Internship & Placement Management System → Company Module

**** ↓ ****

**** Admin Module ****

**** ↓ ****

**** MySQL Database ****

Key Engineering Considerations

- JDBC is used for reliable database connectivity.
- MVC architecture separates user interface, business logic, and database operations.
- Role-based access ensures secure interaction between users.
- Modules are integrated through a centralized database system.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Java Web Application	Manual System	Automation and centralized management	Requires development effort	Selected due to better efficiency
MySQL Database	File Storage	Secure and structured data handling	Requires database setup	Selected for reliability
MVC Architecture	Single-layer Design	Easy maintenance and scalability	Initial complexity	Selected for organized development

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics, Safety and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Digital recruitment process	Reduction of paper-based records	Supports eco-friendly workflow
Ethics	Protection of user information	Secure login and controlled access	Maintains user privacy
Health & Safety	Reduced manual workload	Automated recruitment process	Improves operational efficiency
Security	Data protection	Authentication and database security	Prevents unauthorized access

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Data Loss	Medium	High	Medium	Regular database backup	Low
Unauthorized Access	Medium	High	Medium	Login authentication	Low
System Failure	Low	Medium	Low	Error handling and testing	Low
Database Errors	Medium	Medium	Medium	Validation and testing	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using an incremental development approach where each module was designed, implemented, integrated, and tested separately. The system development involved requirement analysis, database design, user interface development, backend implementation, integration, and testing. The project uses Java-based web technologies with MySQL database connectivity to provide an efficient internship and placement management platform. The major resources used include Java, JSP, Servlets, JDBC, MySQL, Apache Tomcat Server, Eclipse IDE, and MySQL Workbench.

Resource	Purpose	Resource
Java, JSP, Servlets	Backend and web application development	Java, JSP, Servlets
JDBC	Database connectivity	JDBC
MySQL	Data storage and management	MySQL
Apache Tomcat	Application deployment server	Apache Tomcat
Eclipse IDE	Coding and project development	Eclipse IDE

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The system was implemented as a Java-based web application using JSP and Servlets for user interaction and business logic. JDBC was used to establish communication between the application and MySQL database. Apache Tomcat server was used for deploying and testing the web application.

Tool / Software / Equipment	Purpose	Stage Used
Eclipse IDE	Java web application development	Development Stage
Java JDK	Programming environment	Implementation Stage
JSP & Servlets	User interface and server-side processing	Development Stage
JDBC	Database connectivity	Integration Stage
MySQL Workbench	Database design and management	Database Stage
Apache Tomcat	Web application deployment	Testing Stage

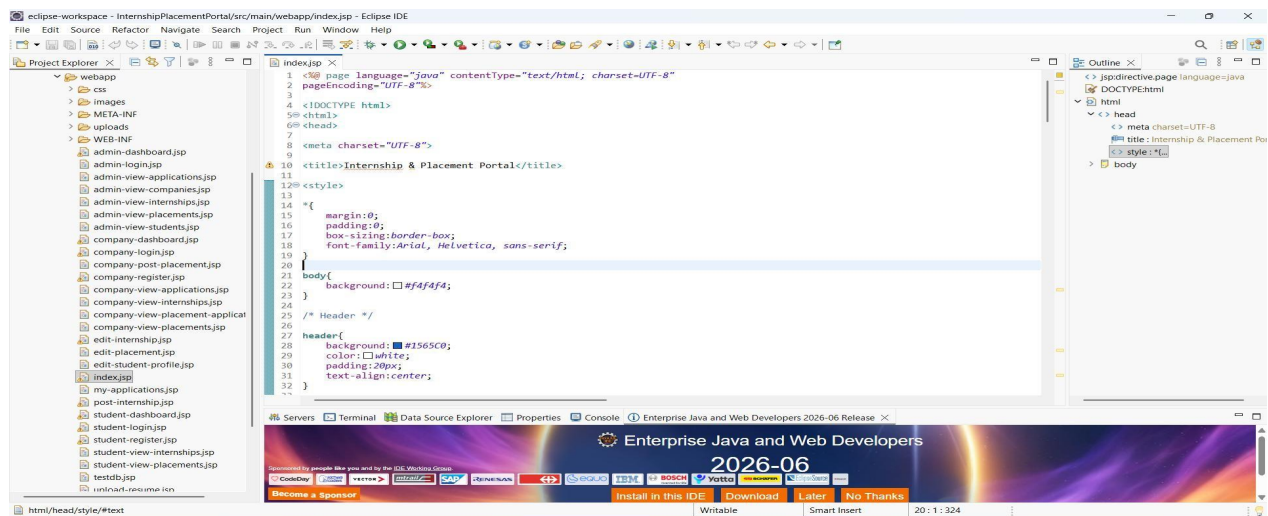


FIG - 03 : Eclipse IDE Implementation

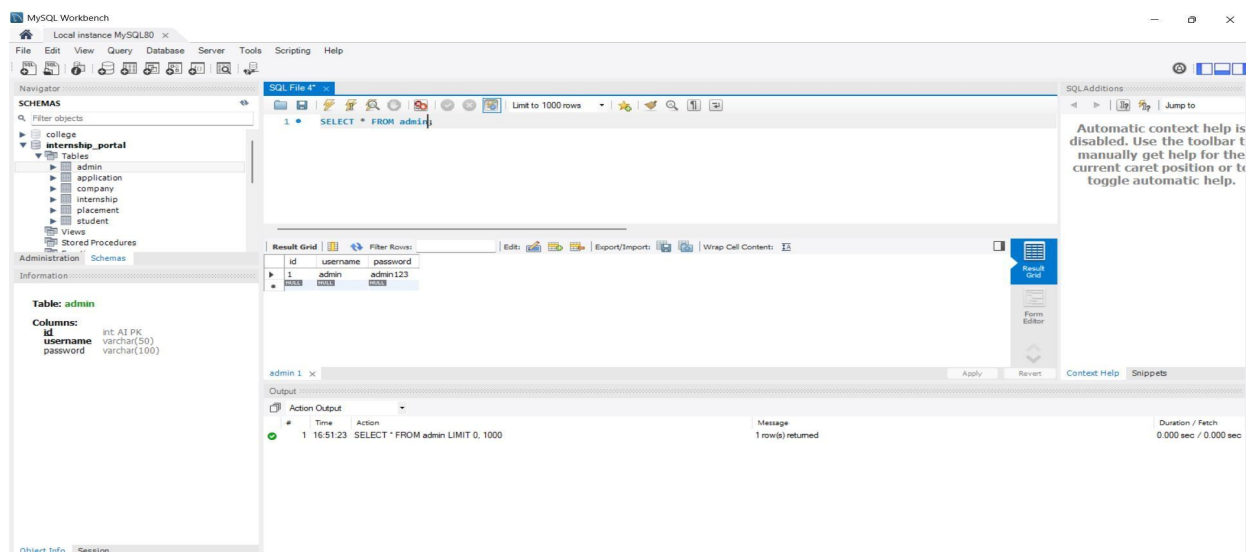


FIG - 04 MySQL Workbench

4.3 Test Plan & Verification

Testing was performed to verify the functionality, reliability, and performance of each module. Functional testing was carried out for student management, company management, internship and placement management, and admin modules.

Test Plan

Requirement	Test Method	Acceptance Criterion
User Login	Functional Testing	Valid users should access the system successfully
Student Registration	Input Validation Testing	Student details should be stored correctly
Company Opportunity Posting	Module Testing	Companies should create and manage opportunities
Application Tracking	Workflow Testing	Application status should update correctly

Verification Results

Specification	Required Value	Achieved Value	Status (Pass/Fail)
User Authentication	Secure login access	Login working successfully	Pass
Database Connectivity	Successful data storage	JDBC connection established	Pass
Application Management	Track applications	Status tracking implemented	Pass
Module Integration	All modules connected	System working successfully	Pass

4.4 Results

The developed system successfully automates internship and placement management activities. The implementation provides separate functionalities for students, companies, and admin. Students can register, upload resumes, view opportunities, apply, and track application status. Companies can post opportunities, view applications, and update candidate status. Admin can manage users and monitor system activities.

Major Results

Module	Result	Module
Student Management	Student registration, profile management, and application tracking implemented	Student Management
Company Management	Company registration and opportunity posting completed	Company Management
Internship & Placement Management	Opportunity management and application processing completed	Internship & Placement Management
Admin Management	User monitoring and system control implemented	Admin Management

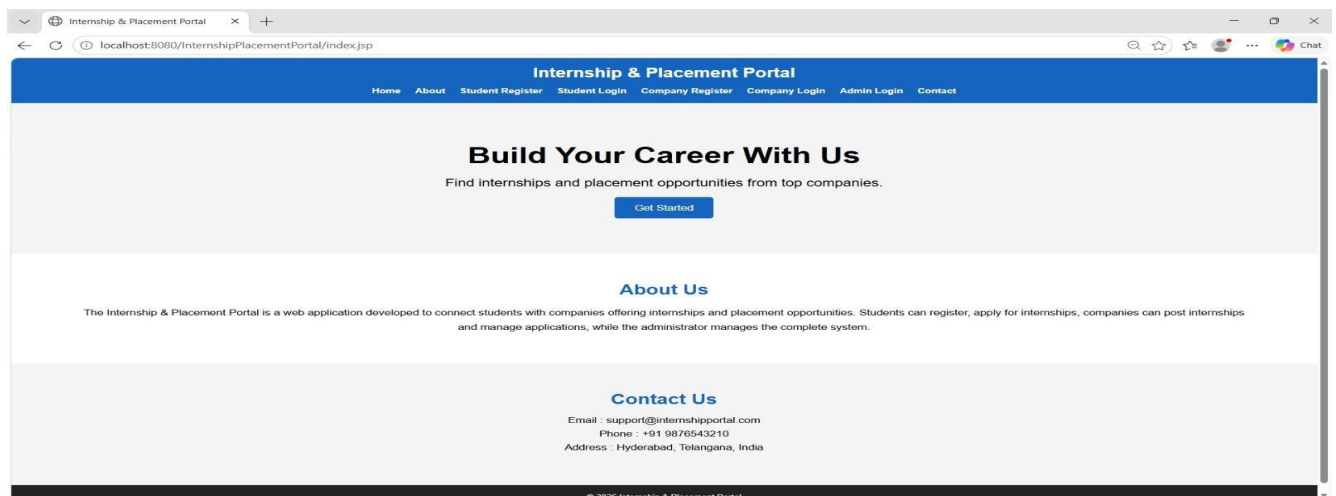


FIG - 05 Internship & Placement Portal Home Page

Student Registration

localhost:5080/InternshipPlacementPortal/student-register.jsp

Internship & Placement Portal

Home Student Login

Student Registration

Full Name

Register Number

Email

Phone Number

Department

Select Department

Year

Select Year

Password

Confirm Password

Register

Already Registered? Login Here

FIG - 06 : Student Registration Page

Student Dashboard

localhost:5080/InternshipPlacementPortal/student-dashboard.jsp

Internship & Placement Portal

Login

Welcome, Tejaswitha

Full Name	Tejaswitha
Register Number	192371657
Email	teja@gmail.com
Phone	7941359130
Department	Computer Science Engineering
Year	IV Year

Edit Profile Upload Resume View Available Internships

View Placement Opportunities My Applications

FIG - 07 : Student Dashboard

Company Registration

localhost:5080/InternshipPlacementPortal/company-register.jsp

Internship & Placement Portal

Home Company Login

Company Registration

Company Name

Email

Phone Number

Location

Website

Password

Register

Already Registered? Login Here

FIG - 08 : Company Registration

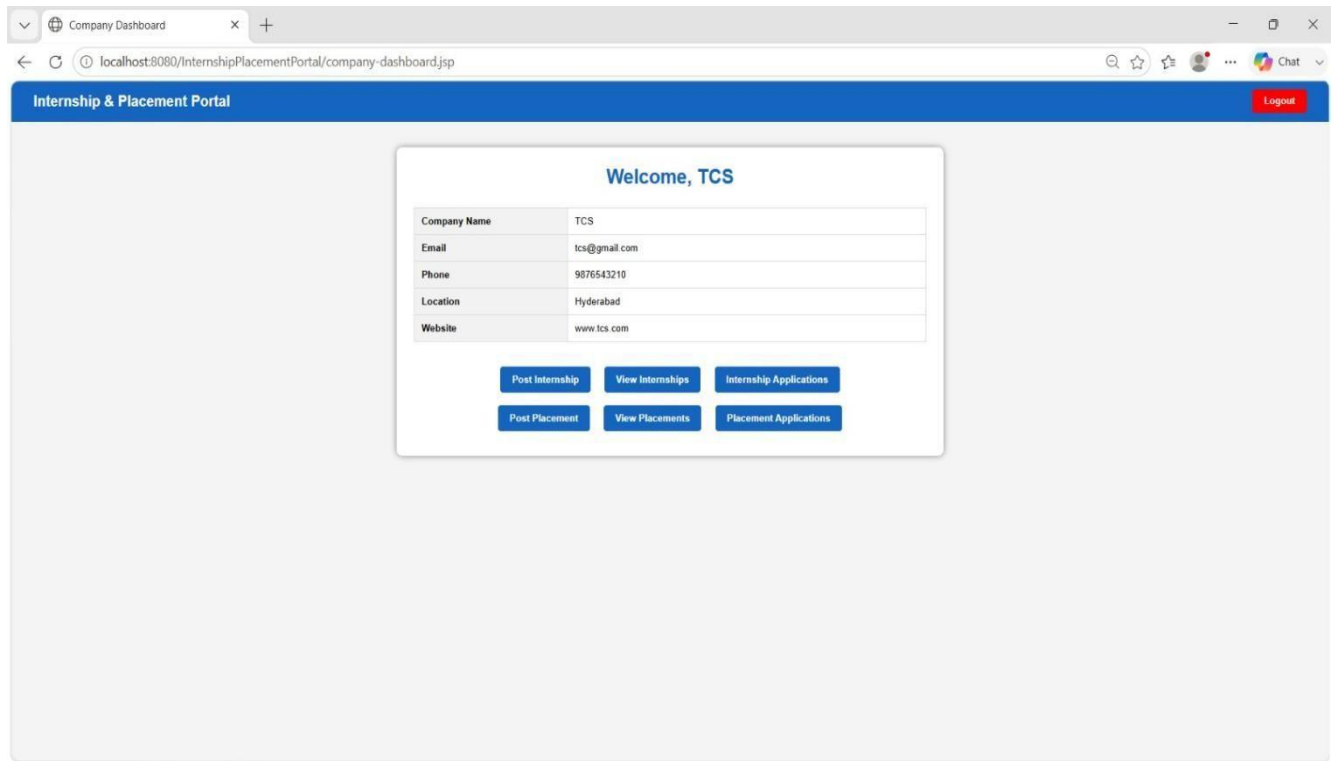


FIG - 09 : Company Dashboard

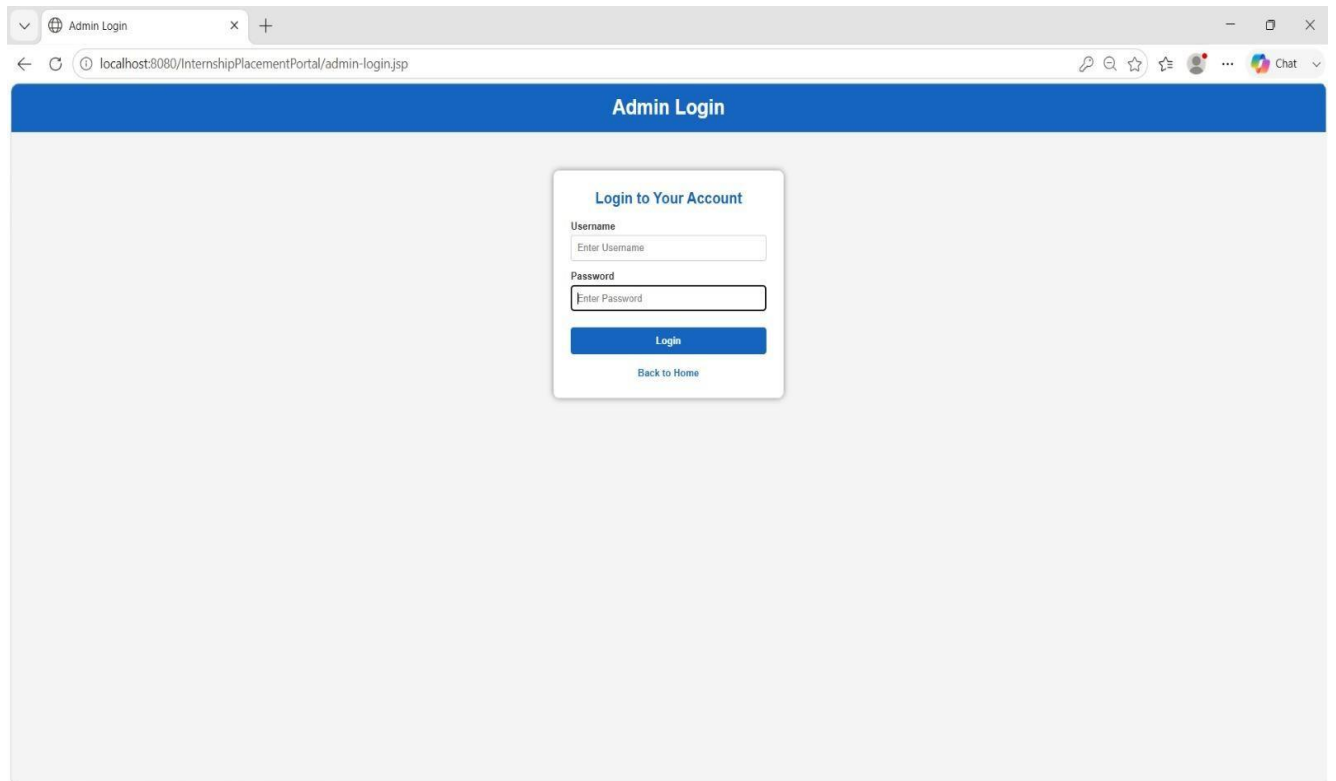


FIG 10 : Admin Login Page

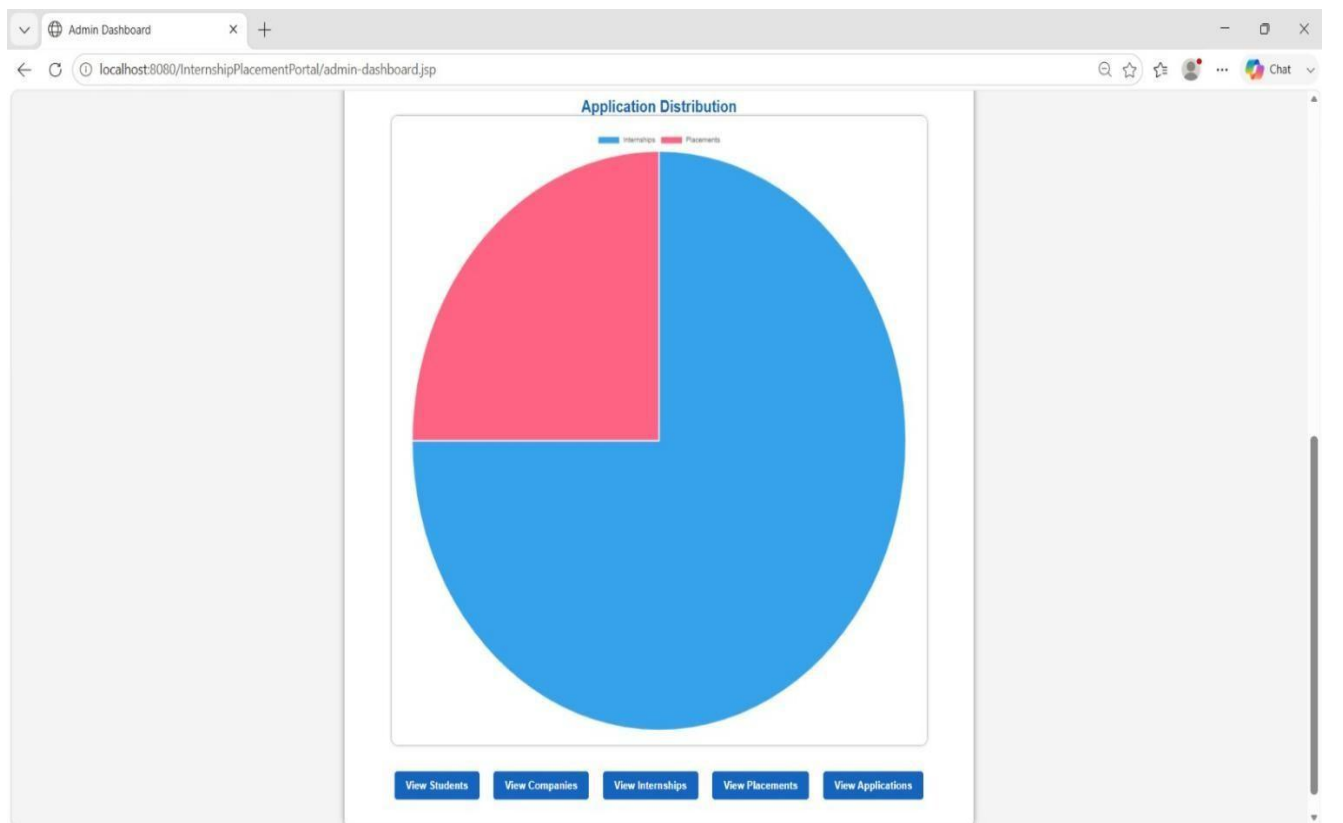
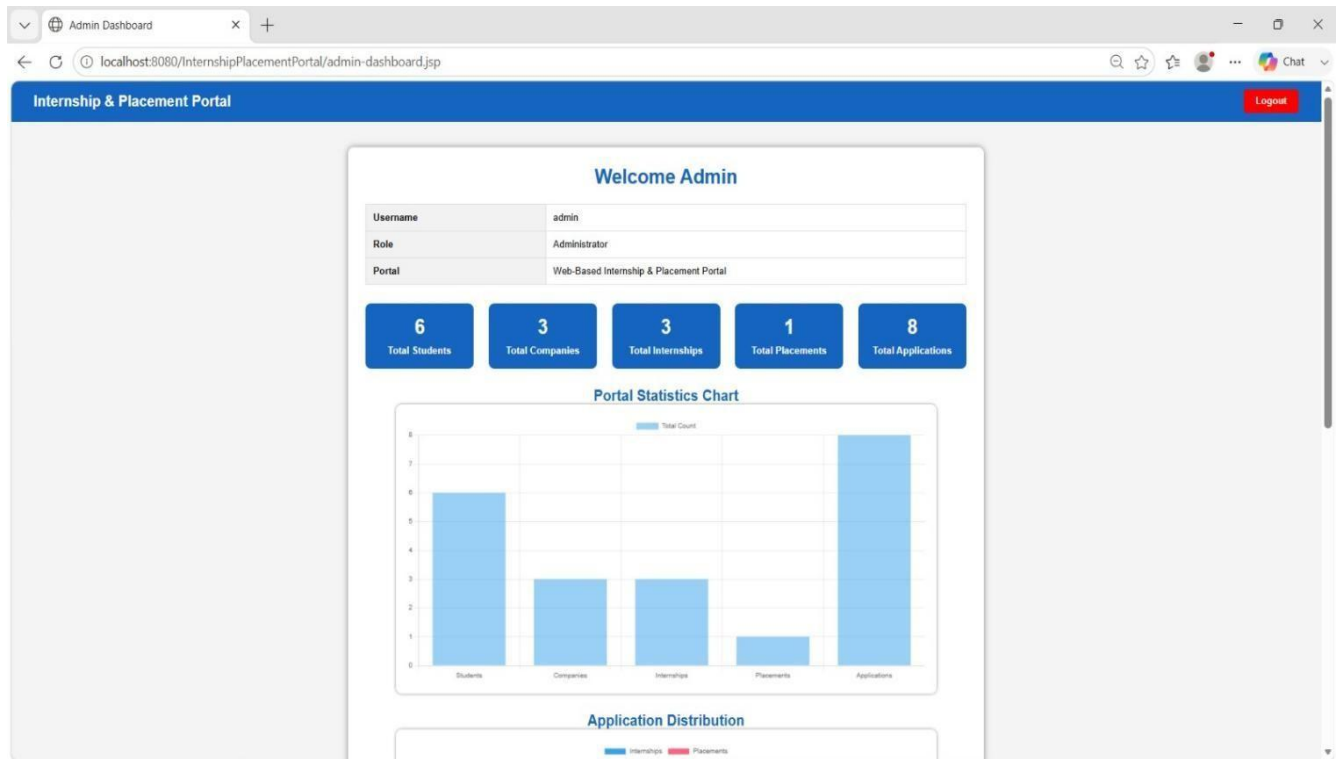


FIG 11 : Admin Dashboard

4.5 Data Analysis & Engineering Judgment

The developed system was evaluated based on functionality, usability, reliability, and database performance. Testing results indicate that all major modules operate successfully and meet the defined requirements. The centralized database approach improves data consistency and reduces manual effort compared to traditional recruitment processes.

The engineering judgment shows that the proposed Java-based web solution provides better scalability, maintainability, and efficiency for managing internship and placement activities. The modular design allows future enhancements such as mobile support, AI-based recommendations, and advanced analytics.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
To design a centralized internship and placement management system	Developed web-based portal with student, company, and admin modules	Fully Achieved
To implement student and company management functionalities	Registration, login, profile management, and opportunity posting modules implemented	Fully Achieved
To manage internship and placement applications efficiently	Application submission and status tracking features implemented	Fully Achieved
To provide secure database-based data management	JDBC connectivity with MySQL database implemented	Fully Achieved
To improve recruitment process efficiency	Automated workflow reduces manual effort	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Provides a centralized platform for students, companies, and admin.
- Reduces manual effort in internship and placement management.
- Maintains structured records using MySQL database.
- Provides easy application tracking and opportunity management.
- Modular design allows future system enhancements.

Limitations:

- The system does not include mobile application support.
- Real-time notification features are not implemented.
- AI-based resume analysis and recommendations are not included.
- System usage is limited to registered users within the institution.

4.8 Project Planning, Roles & Budget

Project Timeline / Milestone Plan

Activity	Duration
Requirement Analysis	Week 1
System Design and Database Design	Week 2
Module Development	Week 3 - Week 6
Integration and Testing	Week 7
Documentation and Final Review	Week 8

Student Contribution

Student	Assigned Responsibility	Actual Contribution
N. Vijetha	Student Management Module	Developed student registration, profile management, resume upload, and application tracking
N. Tejaswitha	Company Management Module	Developed company registration, profile management, and opportunity posting
V. S. Meenakshy	Internship & Placement Management Module	Developed opportunity management and application processing
B. Neha	Admin and System Management Module	Developed admin control, user management, and system monitoring

Project Budget

Item	Estimated Cost	Actual Cost
Software Tools	₹0	₹0
Development Environment	₹0	₹0
Database System	₹0	₹0
Testing and Documentation	₹500	₹500
Total	₹500	₹500

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Web-Based Internship and Placement Management System successfully addresses the challenges involved in managing internship and placement activities through a centralized digital platform. The system provides separate modules for students, companies, internship and placement management, and administration. It automates important activities such as student registration, company management, opportunity posting, application processing, and status tracking. The implementation using Java, JSP, Servlets, JDBC, and MySQL ensures efficient data storage, retrieval, and management. The testing results confirm that the developed system meets the defined objectives by reducing manual effort, improving data organization, and providing a reliable solution for academic recruitment management.

5.2 Future Work

The system can be enhanced further by integrating advanced technologies and additional features. Future improvements include:

- Development of a mobile application for easier access.
- Implementation of AI-based resume analysis and opportunity recommendations.
- Integration of real-time notifications for application updates.
- Addition of advanced analytics and reporting features.
- Cloud deployment for improved scalability and accessibility.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learned the development of Java-based web applications using JSP, Servlets, JDBC, and MySQL.

- Gained practical knowledge of creating database tables, performing CRUD operations, and handling user data securely.
- Understood the importance of modular software design and integration between frontend, backend, and database components.
- Learned debugging techniques for identifying and resolving application errors during development and testing.
- Improved understanding of software development practices such as requirement analysis, implementation, and validation.

Engineering & Professional Skills Developed:

- Improved programming and logical problem-solving abilities through practical implementation.
- Developed skills in backend development, database connectivity, and server-side programming.
- Enhanced knowledge of designing user-friendly interfaces and maintaining structured data flow. Improved testing, debugging, documentation, and project management skills.
- Gained experience in teamwork, task coordination, and following software development methodologies.

Individual Reflection:

- The most difficult engineering problem encountered was managing proper data flow between the Company Management Module and the database. It was solved by implementing JDBC connectivity, validating inputs, and performing module testing.
- A key engineering decision was selecting Java, JSP, Servlets, and MySQL for developing the module because they provide reliable database integration and structured web application development.
- I independently gained knowledge of database operations, server-side programming, and debugging techniques while implementing the company-related functionalities.
- If the project were redesigned, I would include advanced features such as company verification, automated recruiter analytics, and improved dashboard visualization.

REFERENCES

- [1] Oracle Corporation, “Java Documentation,” Oracle. Available:
<https://docs.oracle.com/en/java/>
- [2] Oracle Corporation, “Java Platform, Standard Edition Documentation,” Oracle. Available:
<https://docs.oracle.com/en/java/javase/>
- [3] Oracle Corporation, “JDBC API Documentation,” Oracle. Available:
<https://docs.oracle.com/javase/tutorial/jdbc/>
- [4] Oracle Corporation, “MySQL 8.0 Reference Manual,” Oracle. Available:
<https://dev.mysql.com/doc/>
- [5] Apache Software Foundation, “Apache Tomcat Documentation,” Apache Tomcat. Available:
<https://tomcat.apache.org/>
- [6] Eclipse Foundation, “Eclipse IDE Documentation,” Eclipse Foundation. Available:
<https://www.eclipse.org/documentation/>
- [7] W3C, “HTML Standard Documentation,” World Wide Web Consortium. Available:
<https://www.w3.org/>
- [8] W3C, “CSS Specifications and Standards,” World Wide Web Consortium. Available:
<https://www.w3.org/Style/CSS/>
- [9] I. Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2015.
- [10] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner’s Approach*, 9th Edition, McGraw-Hill Education, 2019.
- [11] M. Fowler, *Patterns of Enterprise Application Architecture*, Addison-Wesley Professional, 2002.
- [12] IEEE Computer Society, “IEEE Standard for Software and System Test Documentation,” IEEE Standards Association.
- [13] IEEE Computer Society, “IEEE Guide to Software Requirements Specifications,” IEEE Standards Association.
- [14] GitHub, “Software Development and Open Source Collaboration Platform.” Available:
<https://github.com/>
- [15] Stack Overflow, “Developer Community and Programming Resources.” Available:
<https://stackoverflow.com/>

APPENDICES

Appendix A – System Architecture and Design Diagrams

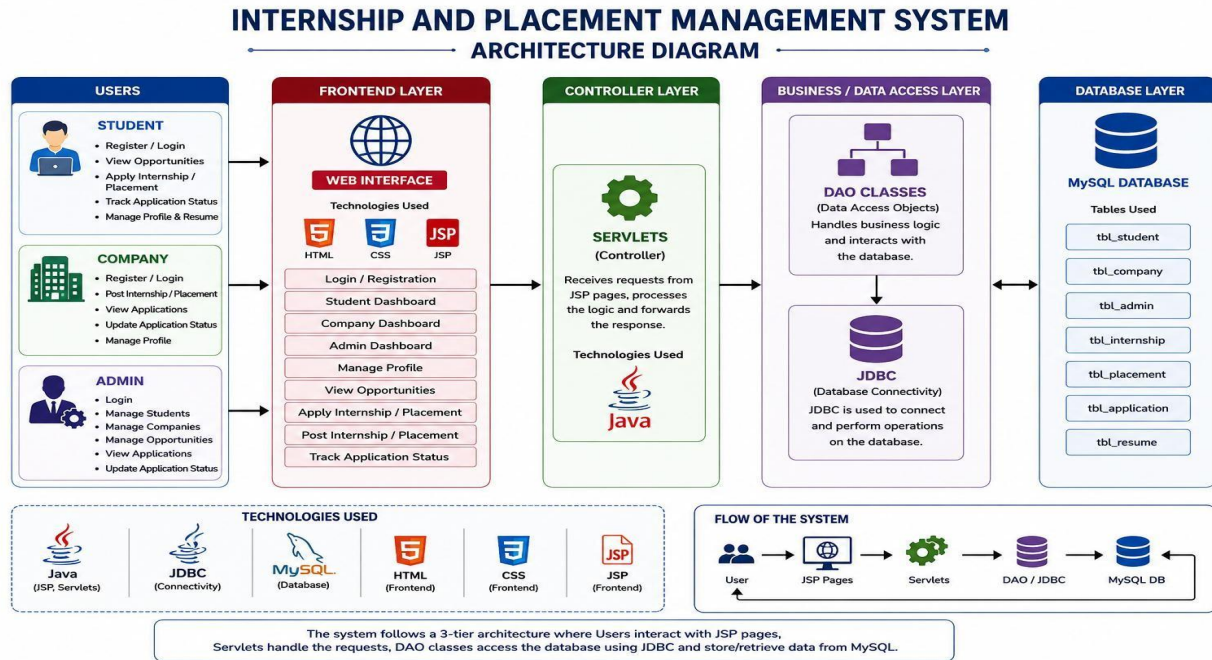


FIG 12 : System Architecture of Internship and Placement Management System

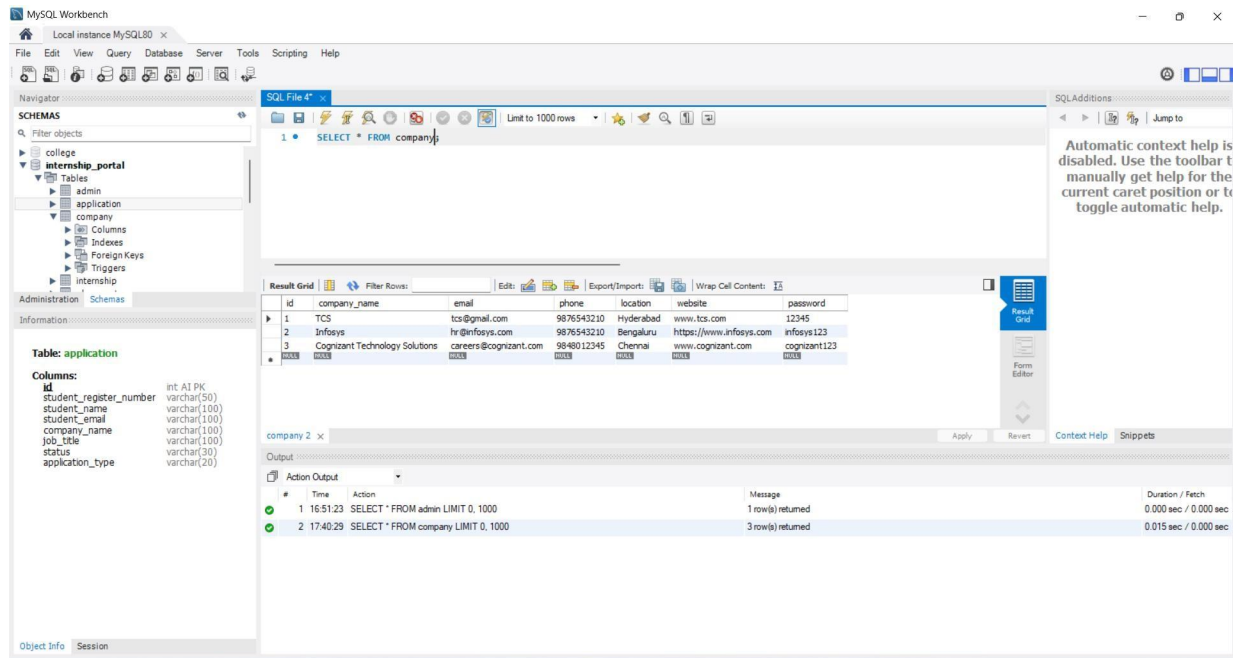


FIG 13 : Database Design of the Proposed System

Appendix B – Source Code / Implementation Evidence

IMPLEMENTATION:

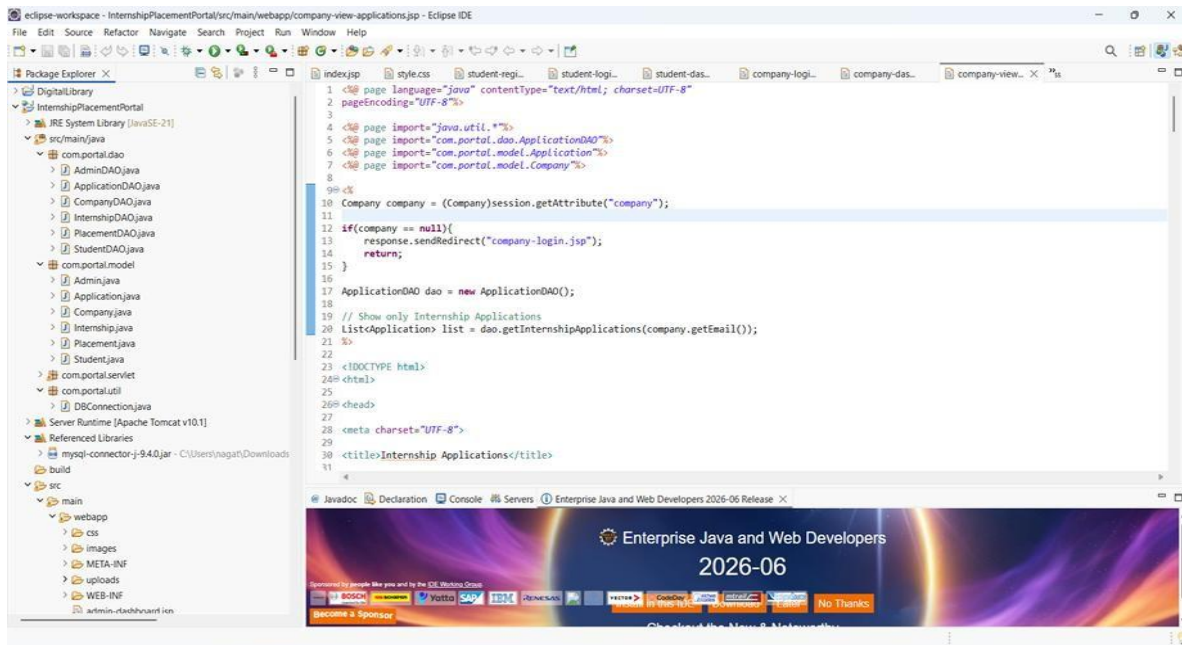


FIG 14 : Web Application Development and Implementation Using Eclipse IDE

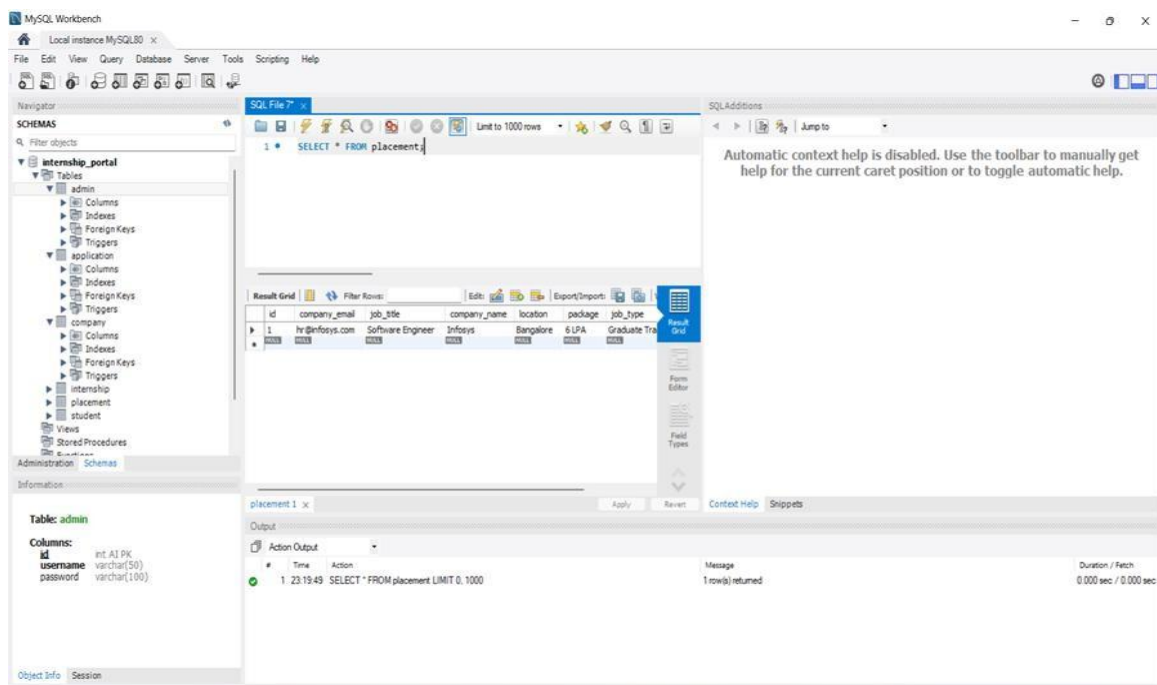


FIG 15 : Database Design and Implementation Using MySQL

Appendix C – Implementation Screenshots (Application)

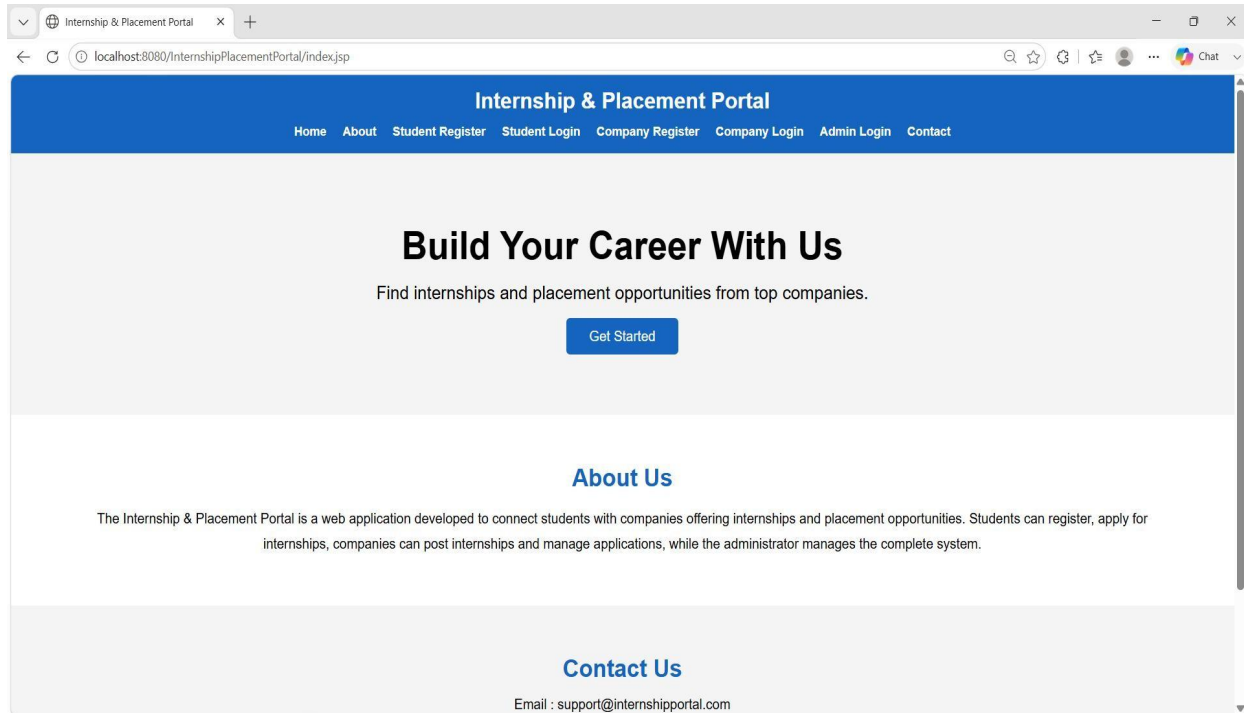


FIG 16 : Internship & Placement Portal

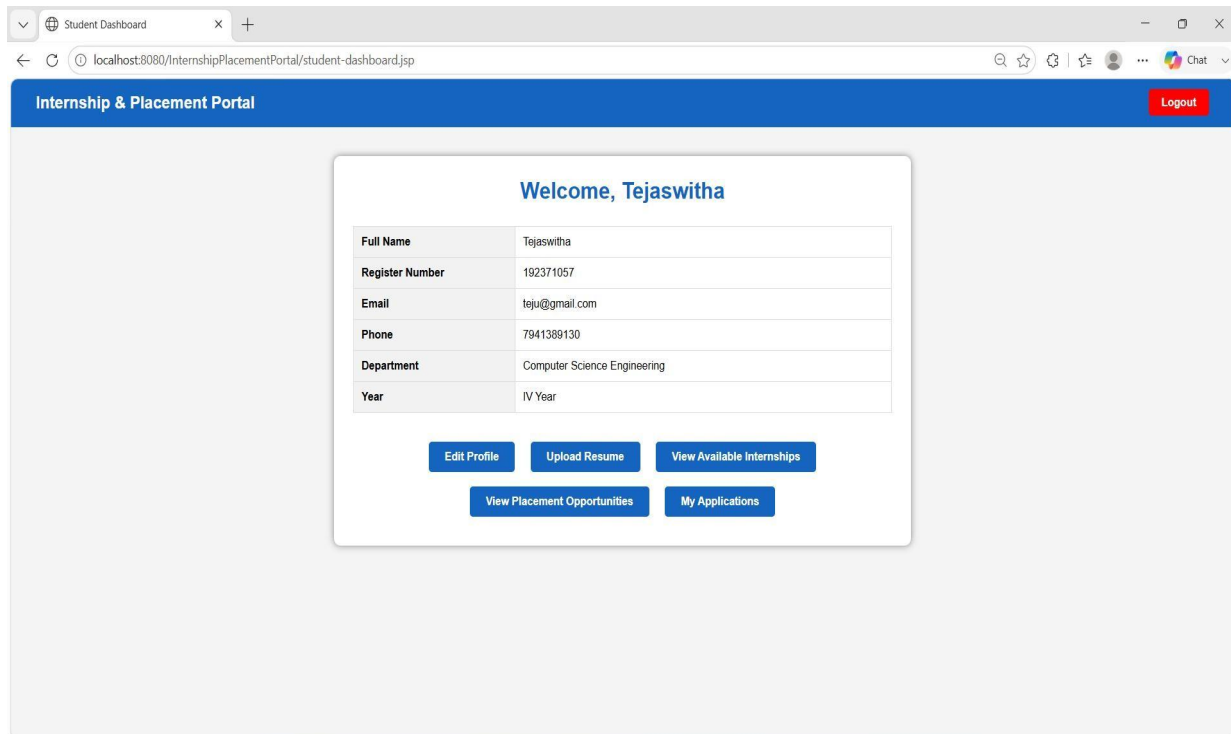


FIG 17 : Student Dashboard

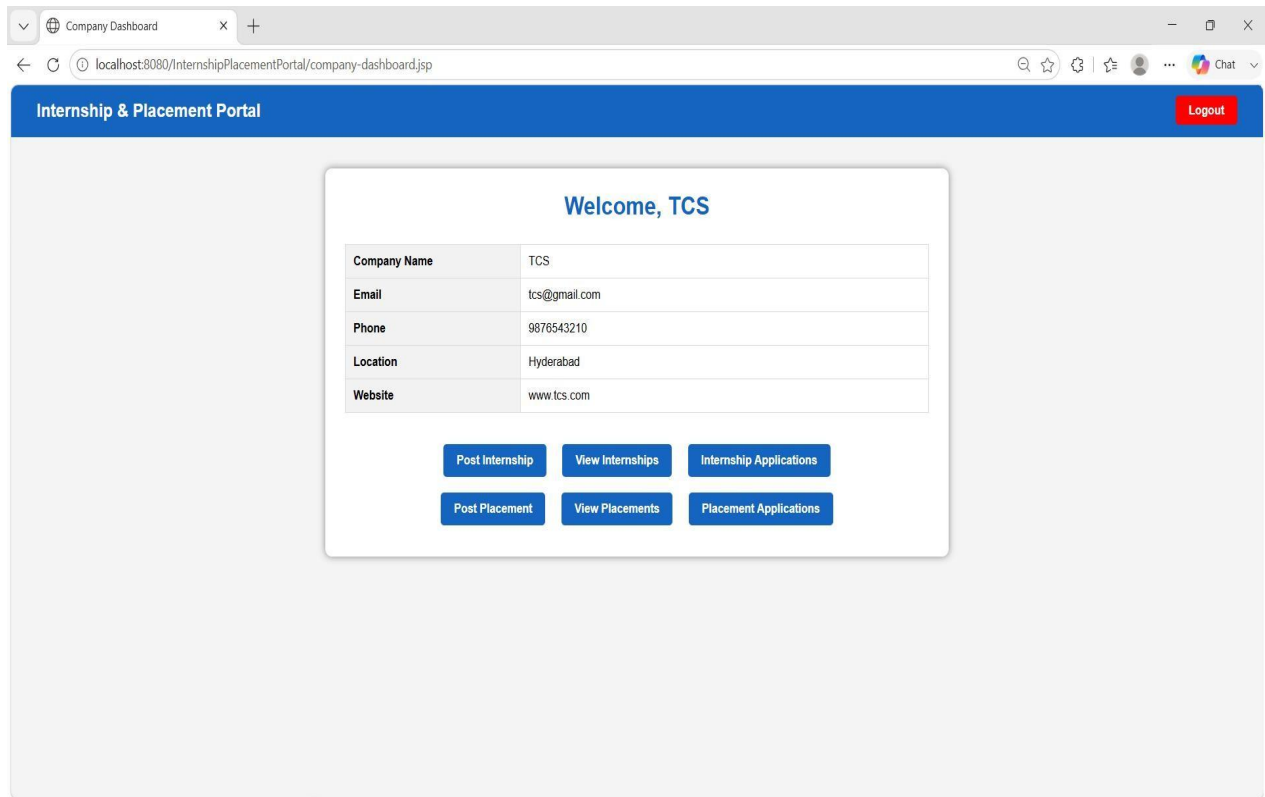


FIG 18 : Company Dashboard

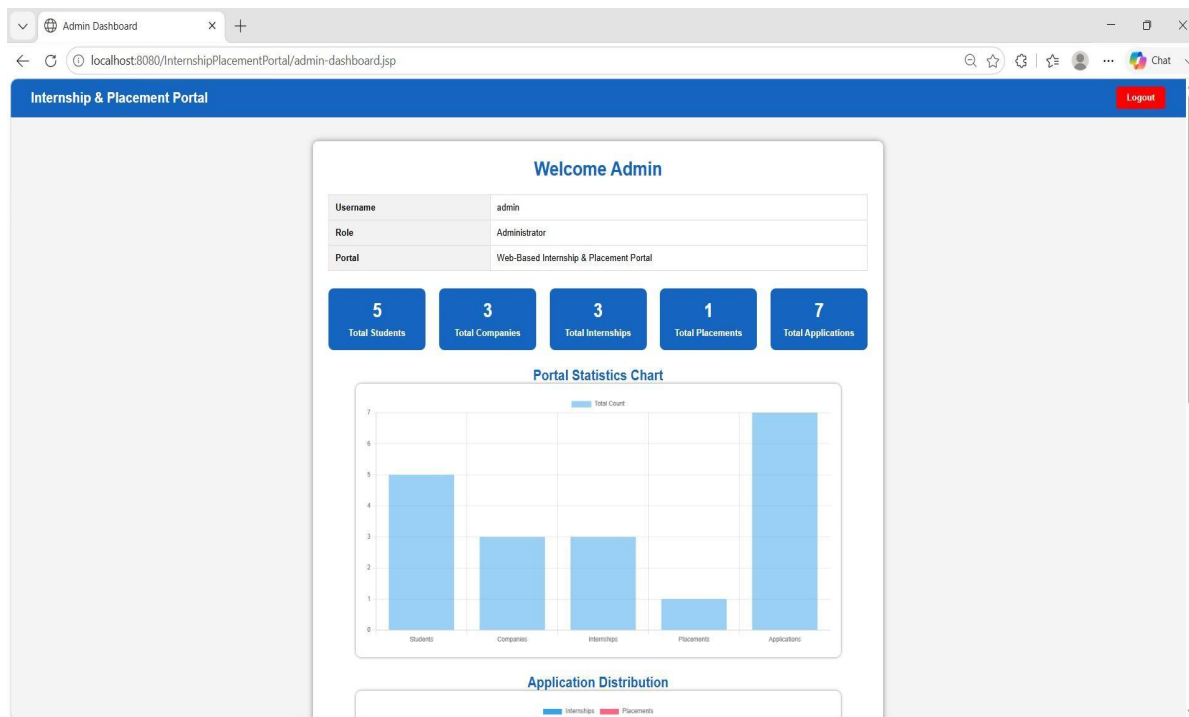


FIG 19 : Admin Dashboard

Appendix D – Testing Evidence

Table D.1: System Testing Results

Test Case	Testing Performed	Expected Result	Status
Database Connectivity	Checked JDBC connection between Java application and MySQL	Database connection established successfully	Pass
User Login	Tested student, company, and admin login functionality	Authorized users can access the system	Pass
Data Management	Tested insertion and retrieval of records	Data stored and displayed correctly	Pass
Application Workflow	Tested internship/placement application process	Applications are recorded successfully	Pass
Module Integration	Tested interaction between all modules	System modules work together properly	Pass

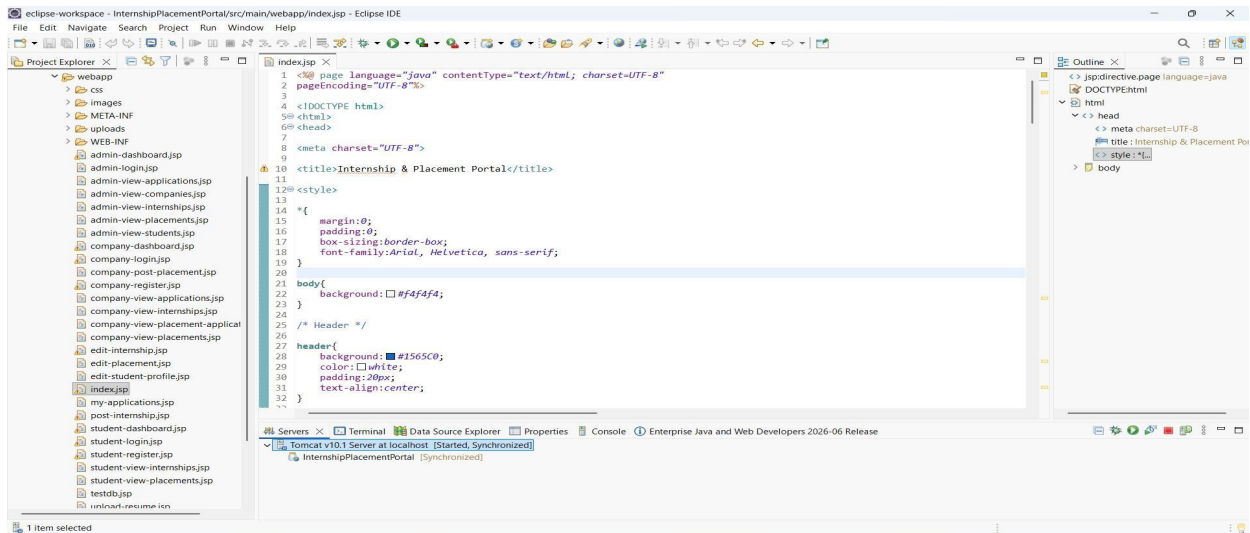


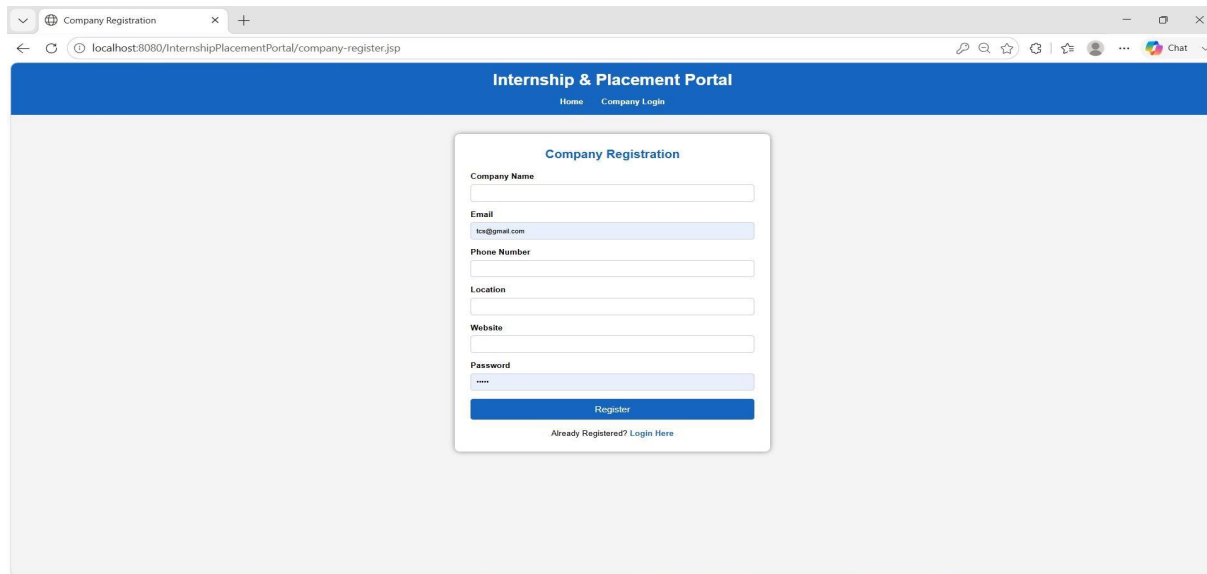
FIG 20 : Server gets Started

Appendix E – Individual Contribution Evidence

Student Name: N.Tejaswitha

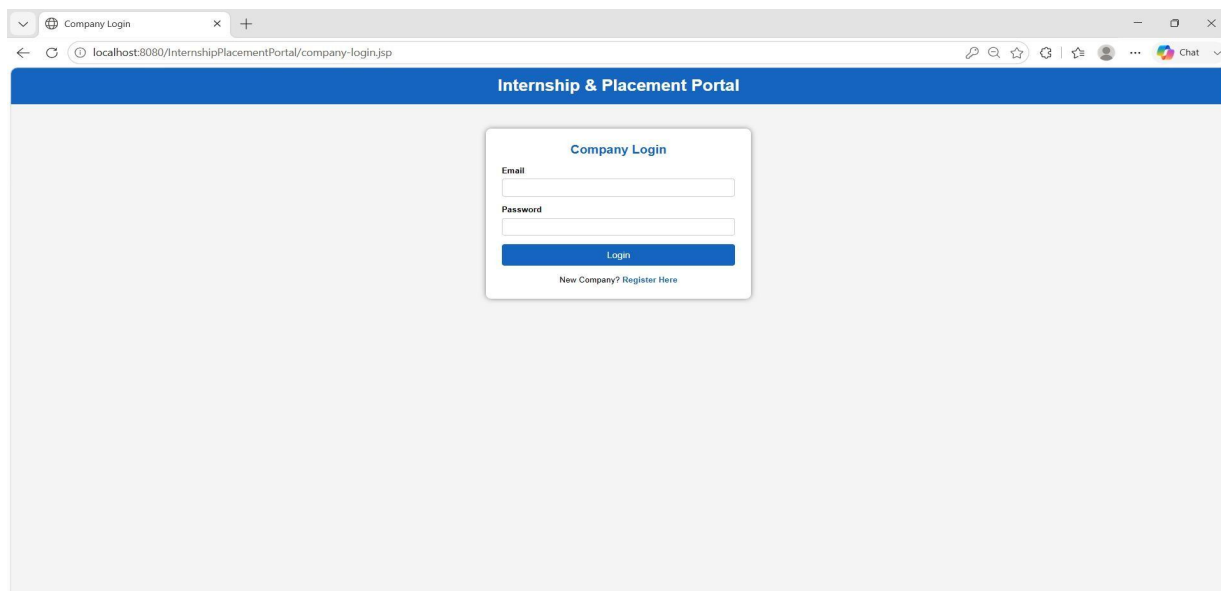
Module: Company Management Module

Evidence :



The screenshot shows a web browser window with the title 'Company Registration'. The address bar shows 'localhost:8080/InternshipPlacementPortal/company-register.jsp'. The page has a blue header with the text 'Internship & Placement Portal' and links for 'Home' and 'Company Login'. The main content area features a 'Company Registration' form with the following fields: Company Name, Email (pre-filled with 'tcs@gmail.com'), Phone Number, Location, Website, and Password (pre-filled with '****'). A blue 'Register' button is at the bottom of the form, with a link 'Already Registered? Login Here' below it.

FIG 21 : Company Registration



The screenshot shows a web browser window with the title 'Company Login'. The address bar shows 'localhost:8080/InternshipPlacementPortal/company-login.jsp'. The page has a blue header with the text 'Internship & Placement Portal'. The main content area features a 'Company Login' form with the following fields: Email and Password. A blue 'Login' button is at the bottom of the form, with a link 'New Company? Register Here' below it.

FIG 22 : Company Login

Company Dashboard

localhost:8080/InternshipPlacementPortal/company-dashboard.jsp

Internship & Placement Portal

Welcome, TCS

Company Name: TCS

Email: tcs@gmail.com

Phone: 9876543210

Location: Hyderabad

Website: www.tcs.com

Post Internship View Internships Internship Applications

Post Placement View Placements Placement Applications

FIG 23 : Company Dashboard

Post Internship

Job Title

Location

Stipend

Duration

Example: 6 Months

Required Skills

Java, JSP, MySQL

Internship Description

Last Date to Apply

dd-mm-yyyy

Post Internship

Back to Dashboard

FIG 24 : Post Internship

View Posted Internships

localhost:8080/InternshipPlacementPortal/view-internships.jsp

View Posted Internships

All Internship Details

ID	Company	Job Title	Location	Stipend	Duration	Skills	Last Date	Edit	Delete
3	Cognizant Technology Solutions	Java Backend Intern	Chennai	₹28,000/month	6 Months	Java, Spring Boot, MySQL, REST API	2026-08-14	Edit	Delete
2	Infosys	Java Full Stack Intern	Bangalore	₹30,000/month	6 Months	Java, JSP, Servlet, MySQL, HTML, CSS	2026-08-28	Edit	Delete
1	TCS	Java Developer Intern	Hyderabad	₹25000/month	6 Months	Java, JSP, MySQL	2026-08-13	Edit	Delete

Back to Dashboard

FIG 25 : View Posted Internships

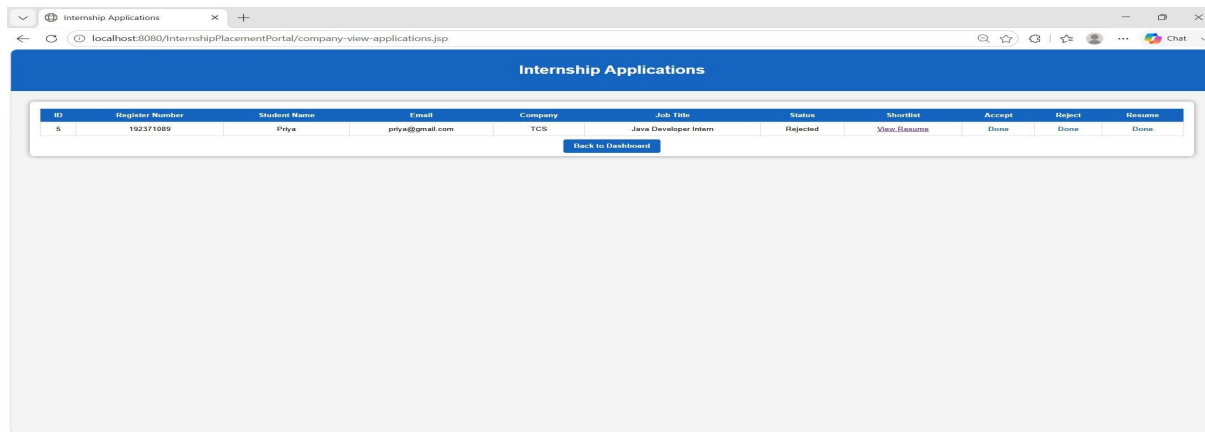


FIG 26 : View Internship Applications

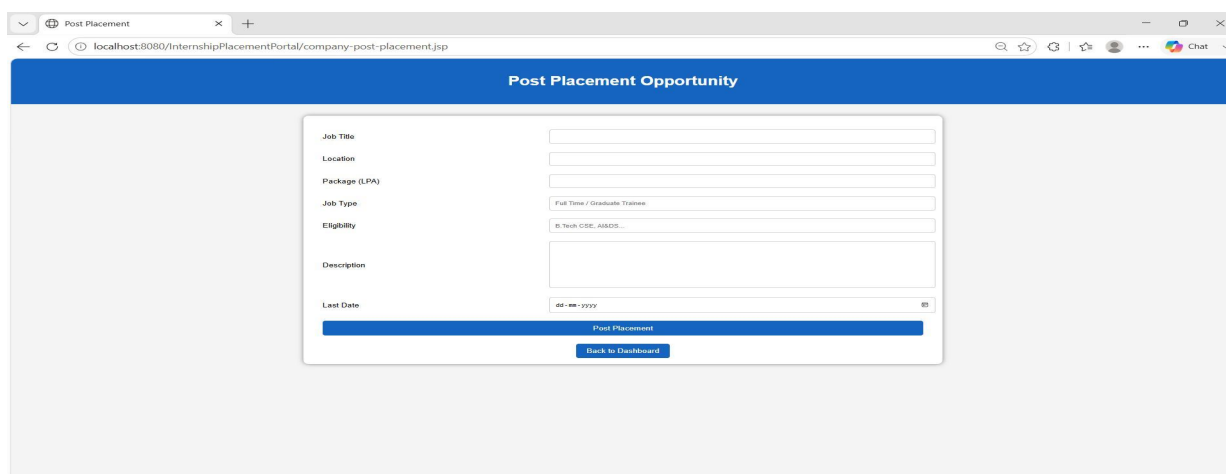


FIG 27 : Post Placement

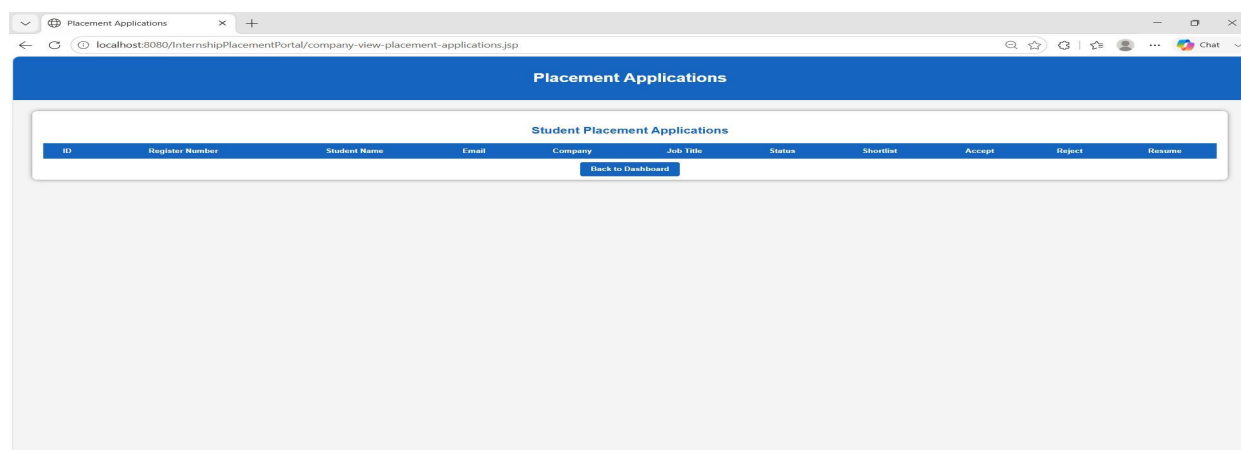


FIG 28 : View Placement Applications

CAPSTONE PROJECT OUTCOME MAPPING SHEET

(To be completed by the department/faculty assessor.)

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1 (1.2, 1.3)
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole Report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8) + Individual Contribution Statement
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Individual Contribution Statement + Ch. 5 (5.3)